

Title

THERMAL ABLATION VERSUS PARATHYROIDECTOMY FOR PRIMARY HYPERPARATHYROIDISM: A SYSTEMATIC REVIEW AND META-ANALYSIS

Authors and Affiliation

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Background

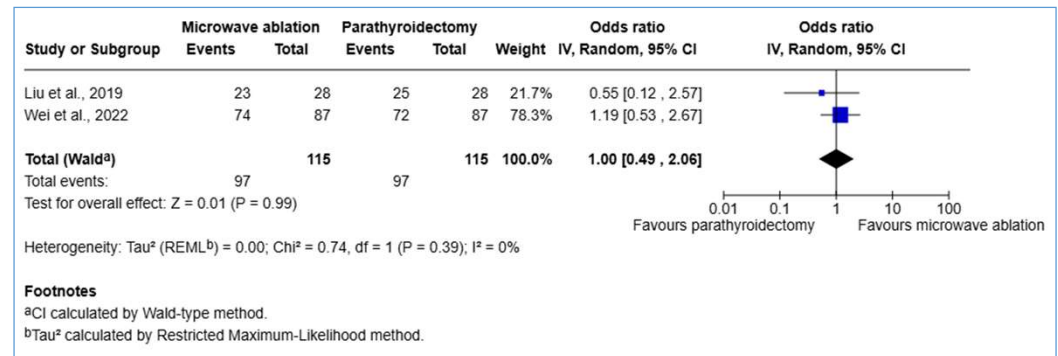
- Primary hyperparathyroidism (pHPT) is an endocrine disorder characterized by autonomous production of parathyroid hormone (PTH), most commonly due to a benign parathyroid adenoma. pHPT precipitates hypercalcemia, leading to nephrolithiasis, osteopenia/osteoporosis, gastrointestinal distress and neuropsychiatric changes.
- Parathyroidectomy (PTX) is the standard treatment for pHPT. Thermal ablation (TA) has emerged as a less invasive alternative.
- Our objective was to **compare the effectiveness and safety of thermal ablation versus PTX in adults with pHPT**, assessing PTH, calcium and phosphorus levels, cure rates, and adverse effects.

Materials and Methods

- We conducted a comprehensive literature search of MEDLINE, EMBASE, and Web of science from inception to March 2025, following the PRISMA guidelines (PROSPERO protocol CRD420251009401).
- We included comparative studies reporting outcomes for adults with pHPT treated with thermal ablation and PTX. Cure was defined as normalization of serum calcium and PTH levels at 6 months follow up.
- Two reviewers screened, extracted data, and assessed risk of bias using ROBINS-I, independently and in duplicate.
- We performed random-effects meta-analysis (RevMan V6) when ≥ 2 studies with comparable interventions and outcome definitions were available. Otherwise, results were synthesized narratively.

Results

- We screened 3,765 records at title/abstract level, assessed 11 full texts, and included 3 comparative retrospective cohort studies conducted in China.
- Mean age range: 53.5-57.7 years, proportion of females range: 65-79%.
- Preoperative serum calcium range: 2.69-2.80 mmol/L, reported adenoma size range: 1.4-2.13 cm.
- One study compared radiofrequency ablation (RFA) vs PTX¹; reporting comparable clinical cure and safety outcomes.
- Meta-analysis was performed for microwave ablation (MWA) vs PTX^{2,3} (2 studies; MWA n=115, PTX n=115). Cure rate, defined as normalization of serum calcium and PTH at 6 months follow-up, was similar between MWA and PTX (odds ratio [OR] 1.00, 95% CI 0.49–2.06, $I^2=0\%$), Figure 1.
- Post-treatment calcium (mean difference [MD] –0.00 mmol/L, 95% CI –0.20 to 0.19, $I^2=93\%$) and phosphorus (MD 0.00 mmol/L, 95% CI –0.05 to 0.05, $I^2=0\%$) were similar. Hypocalcemia was less frequent after MWA (OR 0.33, 95% CI 0.12–0.92, $I^2=24\%$).
- Hoarseness showed a non-significant trend toward being more common with MWA (OR 3.54, 95% CI 0.84–14.94, $I^2=0\%$).
- PTH levels could not be pooled because one study did not report SD.
- The three studies were judged to be at serious risk of bias, primarily due to confounding and participant selection.



Conclusion

- In adults with pHPT, limited comparative evidence suggests MWA achieves **similar cure rates** to PTX with a significantly **reduced risk of hypocalcemia**, but conclusions are constrained by small study numbers and serious **risk of bias**.

Impact Statement

- Demonstrating comparable cure rates to surgery and potentially reduced hypocalcemia introduces new, less invasive options for pHPT patients who refuse or are poor candidates for surgery.
- Larger prospective comparative studies are needed, including direct comparisons across ablation modalities.

References:

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